Riemann surfaces Problem set 2

1. Show that $\mathrm{Aut}(\mathbb{C}^*) \simeq \mathbb{C}^* \rtimes \mathbb{Z}/2\mathbb{Z}$.

2. For $\tau \in \mathbb{H}$, let $L_\tau \subset \mathbb{C}$ be the lattice $L_\tau = \mathbb{Z} \cdot 1 \oplus \mathbb{Z} \cdot \tau$.

(a) Show that any complex torus $X = \mathbb{C}/L$ is biholomorphic to $\mathbb{C}/L_\tau$ for some $\tau \in \mathbb{H}$.

(b) Show that two complex tori of the form $X_i = \mathbb{C}/L_{\tau_i}, i = 1, 2$, where $\tau_1, \tau_2 \in \mathbb{H}$, are biholomorphic if and only if there exists $g \in \mathrm{PSL}(2, \mathbb{Z})$ such that $\tau_2 = g(\tau_1)$.

3. Let $\phi : \mathbb{C} \to \mathbb{C}$ be a homeomorphism and let U be the domain $U = \phi(\{1 < |z| < 2\})$. Show that there is a conformal map $f : U \to A_R$ for some $R > 1$, where $A_R = \{1 < |z| < R\}$.

4. Let $\Gamma < \mathrm{Aut}(\mathbb{H})$ be a discrete, fixed point free subgroup. Show that for any $p \in \mathbb{R} \cup \{\infty\}$, the stabilizer of p in Γ, $\mathrm{Stab}(p) = \{g \in \Gamma \mid g(p) = p\}$, is either trivial or is an infinite cyclic group.

5. Let X be a compact Riemann surface of genus $g \geq 2$ and let $\Gamma < \mathrm{Aut}(\mathbb{H})$ be the group of deck transformations of the universal covering $p : \mathbb{H} \to X$.

a) Show that for any compact $K \subset \mathbb{H}$ there is an $\epsilon > 0$ such that for all $z \in K$, the restriction of p to $B(z, \epsilon)$ is one-to-one.

b) Show that all elements of $\Gamma - \{id\}$ are hyperbolic.

6. Let $R = P/Q \in \mathbb{C}(z) - \mathbb{C}$ be a nonconstant rational function, and let $d = \max(\deg P, \deg Q) \geq 1$. Show that $[\mathbb{C}(z) : \mathbb{C}(R(z))] = d$.

7. Let L be a lattice in $\mathbb{C}$.

a) Show that the sum

$$\mathcal{P}_L(z) := \frac{1}{z^2} + \sum_{\omega \in L - \{0\}} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right)$$

converges uniformly on compacts in $\mathbb{C} - L$ to a function $\mathcal{P}_L$ holomorphic on $\mathbb{C} - L$.

b) Show that the function $\mathcal{P}_L$ is meromorphic on $\mathbb{C}$ with double poles at the points of L.

c) Show that the function $\mathcal{P}'_L$ is an odd, L-periodic function.

d) Show that the function $\mathcal{P}_L$ is an even, L-periodic function.

In view of (c), (d) above, we may identify $\mathcal{P}_L, \mathcal{P}'_L$ with meromorphic functions on the complex torus $\mathbb{C}/L$.

e) Show that the degree of $\mathcal{P}_L$ as a branched covering of $\hat{\mathbb{C}}$ equals 2.

f) Show that $\mathcal{M}(\mathbb{C}/L) = \mathbb{C}(\mathcal{P}_L, \mathcal{P}'_L)$.

Suppose that $L = \mathbb{Z} \cdot \omega_1 \oplus \mathbb{Z} \cdot \omega_2$. Let $\omega_3 = \omega_1 + \omega_2$, and let $e_i = \mathcal{P}_L(\omega_i/2), i = 1, 2, 3$.

g) Show that $\mathcal{P}'_L$ has simple zeroes at the points $\omega_i/2, i = 1, 2, 3$.

h) Show that

$$\mathcal{P}'_L(z)^2 = 4(\mathcal{P}_L(z) - e_1)(\mathcal{P}_L(z) - e_2)(\mathcal{P}_L(z) - e_3)$$

for all $z \in \mathbb{C} - L$.

8. Let X be a compact Riemann surface of genus $g \geq 2$. Show that $\mathrm{Aut}(X)$ is finite.

9. Let X be a Riemann surface such that $\pi_1(X)$ is a non-trivial cyclic group. Show that X is biholomorphic either to $\mathbb{C}^*$ or $\mathbb{D}^*$ or $A_R = \{1 < |z| < R\}$ for some $R > 1$.

10. Show that for any $n \geq 2$, $\mathrm{Aut}(\mathbb{H})$ contains a discrete fixed point free subgroup Γ isomorphic to the free group $\mathbb{F}_n$.